

A Small Exploratory Note on Exact Huber GLMs after Sparsification

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Abstract

While reading *Sparsifying Generalized Linear Models* [1], I tried to understand what happens to exact Huber losses in the high-accuracy refinement part of the paper. The sparsification layer covers Huber-type objectives, but the stated refinement theorem, through Assumption 4.3 on approximate divergence sparsifiability, uses a lower homogeneity parameter $\theta > 1$. Since exact Huber is quadratic near the origin and affine in the tails, it seems to be a boundary case for this uniform local refinement framework. This note records a small local diagnostic check suggesting that the issue is not a lack of structure, but that the useful curvature is active-set dependent and sensitive to active-set switching near the Huber kink.

1 Motivation

My current understanding of [1] is that there are two related but different layers. The first is a sparsification layer: for

$$F(x) = \sum_{i=1}^m f_i(\langle a_i, x \rangle - b_i),$$

the paper gives near-linear size sparsifiers under structural assumptions on the losses. This layer includes Huber-type objectives, in particular the γ_p family, whose $p = 1$ case is exact Huber.

The second is an optimization/refinement layer. In Section 4, the paper uses sparsified local Bregman-divergence models inside an iterative refinement scheme. The main general high-accuracy refinement theorem, Theorem 4.4, assumes Assumption 4.3, including a uniform lower homogeneity parameter $\theta > 1$.

This made me wonder whether the obstacle at the exact Huber endpoint is not the absence of local structure, but rather that the current refinement template may be too uniform to exploit the local quadratic part that is present only on an active set of residuals.

2 Local diagnostic: exact Huber Bregman divergence

Write the scalar Huber loss as

$$H_\delta(t) = \begin{cases} \frac{1}{2}t^2, & |t| \leq \delta, \\ \delta|t| - \frac{1}{2}\delta^2, & |t| > \delta. \end{cases}$$

For a residual y and scalar step s , define

$$D_y(s) = H_\delta(y + s) - H_\delta(y) - H'_\delta(y)s.$$

A direct check gives three regimes.

If both y and $y + s$ stay in the central region $(-\delta, \delta)$, then

$$D_y(s) = \frac{1}{2}s^2.$$

So this part contributes genuine second-order curvature. If both y and $y + s$ stay in the same affine tail, then

$$D_y(s) = 0.$$

So the tail contributes first-order information but no local curvature. The remaining case is crossing one of the kinks $\pm\delta$, where an active-set switching error appears.

For the full GLM objective, at an iterate x , let

$$r_i(x) = \langle a_i, x \rangle - b_i, \quad Q(x) = \{i : |r_i(x)| < \delta\}.$$

Here $Q(x)$ is the active quadratic set. For a small step Δ that does not move many residuals across the kinks, the local model has the form

$$F(x + \Delta) \approx F(x) + \langle \nabla F(x), \Delta \rangle + \frac{1}{2} \sum_{i \in Q(x)} (\langle a_i, \Delta \rangle)^2 + \text{switching error}.$$

Thus, in the unweighted picture, the local curvature is carried by

$$A_{Q(x)}^\top A_{Q(x)}.$$

In a sparsified local model, the corresponding object would be a weighted analogue of this active quadratic curvature. This suggests that exact Huber does have useful local curvature, but it is active-set dependent rather than uniform over all residuals.

3 Relation to nearby work

There are several nearby tools, but I do not see an immediate way to plug them directly into the refinement theorem of [1]. Sketching and sampling methods for robust regression and M-estimation handle Huber-type losses in approximation settings [2, 3], which suggests that Huber objectives are not intrinsically resistant to sampling. However, the question here is more specific: after a near-linear sparsification step, can one still run a high-accuracy local refinement scheme that exploits the exact Huber active-set geometry?

Semismooth Newton methods for Huber regression explicitly use the piecewise smooth structure of the loss [4]. Smooth surrogates such as pseudo-Huber also fit more naturally into self-concordance-style curvature frameworks [5, 6]. But these do not directly answer the boundary case I am trying to understand, because exact Huber has genuinely affine tails and kink-switching behavior.

Thus the question I am focusing on is narrower: it is not whether Huber regression can be optimized or sampled in general, but whether the near-linear sparsification framework can be combined with a high-accuracy local refinement step for the exact, piecewise-smooth Huber endpoint.

4 What I found and where I got stuck

My preliminary conclusion is:

- exact Huber locally decomposes into a quadratic part on $Q(x)$ and affine pieces outside $Q(x)$;
- in the unweighted picture, the relevant curvature for a local second-order step is carried by $A_{Q(x)}^\top A_{Q(x)}$, while a sparsified local model would need to preserve a weighted analogue of this active curvature;
- a sparsified refinement step would need to preserve this active quadratic curvature while controlling switching near $\pm\delta$.

This suggests a possible local model

$$\min_{\Delta} \left\{ \langle \nabla F(x), \Delta \rangle + \frac{1}{2} \sum_{i \in Q(x)} (\langle a_i, \Delta \rangle)^2 + \lambda \|\Delta\|^2 \right\},$$

possibly with a trust-region constraint to limit active-set changes. A semismooth Newton variant might be another way to express the same idea.

I do not know whether this can be made compatible with the sparsification guarantees in [1]. In particular, I am unsure whether the switching error can be controlled without losing the near-linear sparsity advantage.

5 Question

My main question is:

For exact Huber losses, should the apparent endpoint gap between the sparsification result and the stated high-accuracy refinement applications be viewed mainly as a limitation of the current uniform local divergence-sparsifiability template, or is there a more fundamental barrier after sparsification?

More concretely, is it reasonable to try a local refinement step based on the active quadratic set $Q(x)$, perhaps using a semismooth, proximal, or trust-region model? Or does active-set switching near the Huber kink make such an approach unlikely to work?

References

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